Anatomy of a 2–3σ Hint

Bobby Morong[1]

[1] *Independent researcher*

(Dated: August 2026)

Abstract

We independently reimplement the published compressed likelihoods behind the DESI DR2 preference for evolving dark energy and audit the result with selection-aware simulations, direct time-variation tests, held-out prediction, supernova-calibration sensitivity, and full-Boltzmann checks. The compressed DESI+CMB likelihood gives $\Delta\chi^2 = -8.4575$ for CPL relative to ΛCDM; 70 of 5,000 fitted-ΛCDM mocks are at least as extreme, $p = 0.0140$. The largest leave-one-tracer influence is LRG2 at $z = 0.706$, $M = 4.1249$, with an unconditional selected tail of 78/5,000, $p = 0.0156$. Conditional on null mocks with a global preference at least as strong as observed, however, 34/70 have an equal-or-larger maximum, $p_{\rm cond} = 0.486$: LRG2 is load-bearing, but its localization is not independent evidence. A direct constant-w-versus-CPL calibration gives $T = 7.8506$, 30/5,000, $p = 0.0060$ (2.75σ); without LRG2 it falls to $T = 3.4263$, 349/5,000, $p = 0.0698$ (1.81σ). A targeted held-out ΛCDM prediction gives joint LRG2 $p = 0.0408$, driven by isotropic distance rather than an unusual Alcock--Paczynski ratio. A free low-redshift DES5Y magnitude offset reduces the corresponding supernova preference to noise-compatible levels, as a degeneracy test rather than a calibration diagnosis. CAMB and a PR3-native full-CMB variant preserve the LRG2 sensitivity. The compressed likelihood therefore contains a calibrated, CPL-specific 2--3σ hint, but not a discovery. Independent human clean-room execution remains pending.

I. MOTIVATION AND SCOPE

The DESI DR2 BAO analysis (arXiv:2503.14738) reports a preference for a time-varying dark-energy equation of state reaching 3.1σ with full CMB data and 2.8–4.2σ with supernova compilations — the most significant challenge to ΛCDM in a generation, if it holds. Claims of this magnitude attract two failure modes: uncritical amplification and uncritical dismissal. This project attempts a third path: full independent reimplementation followed by systematic attempts to break the result, with every finding published regardless of direction.

Scope, stated precisely: this is an independent implementation of the *published compressed likelihoods* — DESI DR2 BAO summary values, Pantheon+ and DES-SN5YR distance moduli with full covariances, and the DR2 paper's own compressed early-CMB prior (its eqs. 35–36). It is not a raw-data reanalysis. In addition to the compressed branch, a full-CMB variant (Planck 2018 native low-ℓ TT/EE + plik-lite TTTEEE + lensing, via cobaya+CAMB) is included; DESI's own analysis uses PR4 CamSpec and PR4/ACT lensing, so this variant tests the lensing-inclusive structure of the result, not its exact published numbers. Four AI systems performed software, statistics, or manuscript checks during development; these are not independent scientific peer review. One re-executed the pipeline and matched all χ^2 values to four decimals. A clean-room human rerun remains the outstanding verification step.

II. DATA AND PROVENANCE

All inputs are public and primary-sourced under a fixed rule: no number enters the pipeline from memory. DESI DR2 BAO values (7 tracers, correlated D_M/D_H pairs plus BGS D_V) and the compressed CMB prior were machine-extracted from the arXiv HTML of the DR2 paper. Pantheon+ (1,701 light curves; 1,580 after standard cuts; full STAT+SYS covariance) from the official PantheonPlusSH0ES release. DES-SN5YR from the official repository's 2025 Dovekie-recalibrated release (N = 1,820; inverse covariance unpacked per the repository's own likelihood code; external low-z sample identified by ID-SURVEY). eBOSS DR16 LRG consensus values from the official SDSS files in the cobaya bao_data release. DESI DR1 values from arXiv:2404.03002. BBN and θ_* priors from DR2 eqs. 14 and 16; w(z) priors ($w_0 \in$ U[-3,1], $w_a \in$ U[-3,2], $w_0 + w_a < 0$) verbatim from the paper.

III. METHODS

Likelihoods are Gaussian in the published summary statistics (mirroring the official compressed treatment), with supernova magnitude offsets marginalized analytically and, where used, subsample calibration offsets marginalized jointly. Late-time distances use a validated fast integrator (relative error $< 3 \times 10^{-7}$, verified by external audit). Early-universe quantities (r_drag, θ_*) use, in the production pipeline, a calibrated surrogate — standard fitting formulas pinned to the paper's own r_d formula and CMB column, a labeling adopted after audit — and, decisively, full CAMB Boltzmann computations (RECFAST recombination, PPF dark energy) for verification. Two validation gates preceded all dark-energy fits and caught two real implementation bugs before any result existed. The ΛCDM-versus-w_0w_a

statistic is calibrated with 5,000 ΛCDM mocks. The direct one-extra-parameter test of time variation compares wCDM to CPL w_0w_aCDM and is calibrated with 5,000 mocks at the observed best-fit constant-w null, both with and without LRG2. The official w_0, w_a, and $w_0 + w_a$ prior domain is enforced in every fit.

IV. REPRODUCTION

- **BAO-alone ΛCDM: This work:** $\Omega_m = 0.2973 \pm 0.0083$, h · r_d = 101.55 ± 0.72; **Published:** 0.2975 ± 0.0086, 101.54 ± 0.73
- **Pantheon+-alone ΛCDM: This work:** $\Omega_m = 0.3326 \pm 0.0180$ (N=1580); **Published:** 0.334 ± 0.018
- **$\Delta\chi^2$: DESI / +Pantheon+ / +cCMB: This work:** -4.7 / -5.1 / -8.5; **Published:** -4.7 / -4.9 / -8.0
- **w_0, w_a (DESI+cCMB, MAP): This work:** -0.44, -1.68 (H0 = 63.6); **Published:** -0.42, -1.75 (63.6)
- **w_0, w_a (posterior 68%): This work:** -0.45 +0.24/-0.23, -1.66 +0.65/-0.71; **Published:** -0.42 ± 0.21, -1.75 ± 0.58
- **w_0--w_a posterior correlation: This work:** -0.978; **Published:** -0.975
- **ΔDIC (DESI+cCMB): This work:** -4.96 (MAP plug-in; p_D = 4.75/3.0); **Published:** -4.4

An AI software audit re-ran this pipeline and confirmed all χ^2 values to four decimals, verified the minima against differential evolution, and confirmed covariance positive-definiteness. This is a software check, not independent scientific review.

V. INFLUENCE DIAGNOSTICS: WHERE THE PREFERENCE LIVES

Leave-one-tracer-out refits of DESI+cCMB show the preference is not distributed: dropping any of six tracers leaves 2.3–2.6σ (removing BGS or Lyα *strengthens* it), while dropping LRG2 (z = 0.706) collapses it to 1.6σ. The χ^2 decomposition attributes the w_0w_a improvement to LRG2 (-4.1) and the CMB geometric anchor (-3.9), with LRG1 contributing -1.2, all others approximately 0, and Lyα opposing at +1.0. Within LRG2, the radial measurement carries more model leverage: retaining only D_H/r_d gives $\Delta\chi^2$ = -6.5, only D_M/r_d gives -4.9, neither -4.3. The selection-calibrated statistic I_j = $\Delta\chi^2$_-j - $\Delta\chi^2$_full and M = max_j I_j gives M_obs = 4.125 at LRG2. Repeating the full fit and all seven deletion fits inside each of 5,000 ΛCDM mocks yields an unconditional selected tail of 78/5,000, p = 0.0156 [0.0124, 0.0194]. Conditioning on a global preference at least as strong as observed leaves 70 mocks, of which 34 meet or exceed M_obs: p_cond = 0.486 [0.364, 0.608]. Thus LRG2's leverage is a valid description of where the observed fit changes, but it is not an additional anomaly beyond the rarity of the global fit.

A targeted held-out check fits ΛCDM to DESI+cCMB without LRG2, propagates the fitted-parameter covariance, and then predicts the excluded (D_M/r_d, D_H/r_d) vector. The prediction is (17.530, 20.066), compared with (17.351, 19.455) observed. Repeating the full exclusion, fit, prediction, and score in 5,000 fitted-ΛCDM mocks gives a joint tail of 204/5,000, p = 0.0408 [0.0355, 0.0467] (2.05σ). Its decomposition changes the earlier interpretation: the isotropic distance D_V/r_d is low (p = 0.0204), but the Alcock–Paczynski ratio D_M/D_H is unexceptional (p = 0.374). LRG2's radial coordinate carries more leverage for the $w_0 w_a$ model comparison, but the held-out residual is not an anomalous anisotropy. Because LRG2 was selected after inspection, these targeted p-values are diagnostic and do not replace the selected-null calibration above. Shifting LRG2 halfway toward ΛCDM lowers the global preference below 2σ; inflating only its errors degrades the signal smoothly, indicating statistical leverage rather than numerical pathology. On the supernova side, the significance of DESI+Pantheon+ falls monotonically from 1.8σ to 1.3σ as low-z cuts rise from z > 0.01 to z > 0.1, with w_0 pinned near -0.89 throughout.

VI. THE Z APPROXIMATELY 0.7 MEASUREMENT ACROSS EXPERIMENTS

Three analyses have measured BAO at this redshift. Against the ΛCDM best-fit prediction: eBOSS DR16 (z = 0.698; partially correlated with DESI in sample and footprint) deviates jointly by 0.83σ, with D_M *high* if anything; DESI DR1 by 1.40σ, driven by low D_M; DESI DR2 by 1.51σ, driven by low D_H. The anomaly changed observable between DESI's own releases. The swap test quantifies the consequence: with DR2's LRG2 the preference is 2.4σ; with DR1's, 1.9σ; with eBOSS's, 1.6σ — numerically identical to having no z approximately 0.7 measurement at all. Balanced reading: DR2's measurement is 1.7–1.8× more precise than either comparator and may be resolving what noisier data could not;

DESI's blinded internal-consistency checks passed; eBOSS could not strongly confirm a real signal at its precision, and the footprints partially overlap. But a discovery narrative must carry both facts: the signature is unstable across releases, and the earlier partially overlapping measurement at this redshift is ΛCDM-consistent (the swap is a comparator test, not an independent non-confirmation). Reproducing LRG2 from its official likelihood surface is the single most decisive open test.

VII. SUPERNOVA CALIBRATION

On the Dovekie-recalibrated DES5Y release, DESI+DES5Y (no CMB) prefers $w_0 w_a$ at 2.2σ ($\Delta\chi^2$ = -7.3) — against -13.6 (3.3σ) published with the 2024 release. The updated Dovekie release changed cross-calibration, SALT training, scatter modeling, bias simulations, and sample treatment together; the smaller preference coincides with the new release, but attribution among those changes is not established (the same machinery reproduces Pantheon+ at -5.1 vs -4.9). As a sensitivity test (not a diagnosis — a free offset can absorb calibration, selection, population evolution, survey differences, or genuine cosmological curvature alike), freeing one magnitude offset on the 197 external low-z supernovae returns D = +0.037 mag under ΛCDM — matching Efstathiou's ~0.04 mag estimate from Pantheon+/DES5Y cross-matching — and the preference collapses to 0.2σ, with w_0 = -0.99. With the CMB anchor attached, the intercept still reduces 2.9σ to 1.3σ (D = +0.036). Counter-arguments are part of the record: the DES team reproduces the offset but attributes much of it to improved intrinsic-scatter and host-galaxy modeling (reverting those assumptions moves their result $3.9\sigma \rightarrow 3.3\sigma$, not to zero); the intercept is a post-hoc test motivated by a critic and may double-count covariance systematics; and per Cortês & Liddle, the overlapping supernova compilations must never be averaged.

VIII. ROBUSTNESS OF THE STATISTICAL FRAMEWORK

The preference is not unique to CPL: JBP and logarithmic w(z) families find similar improvements. That parameterization swap, however, is not itself a calibrated test that w changes with time. We therefore test the nested CPL question directly, comparing constant-w CDM to $w_0 w_a$CDM with only w_a added. For DESI+cCMB, T = $\chi^2_{\rm min}$(wCDM)

- $\chi^2_{\min}(w_0 w_a\mathrm{CDM})$ = 7.8506. In 5,000 mocks generated at the observed best-fit constant-w model, 30 are at least as large: p = 0.0060 [0.0041, 0.0086], or 2.75σ; the one-degree-of-freedom Wilks value is 2.80σ. No mock fit failed or reached the prior boundary. After removing LRG2, T falls to 3.4263, with 349/5,000 exceedances: p = 0.0698 [0.0629, 0.0772], or 1.81σ. Thus the compressed data contain a direct, calibrated, CPL-specific preference for time variation, but most of its strength is not retained without LRG2. This does not establish arbitrary time dependence or a physical dark-energy field.

Separately, the 5,000-mock ΛCDM-versus-$w_0 w_a$ global calibration yields 70 exceedances of the unrounded observed $\Delta\chi^2$ = -8.4575: p = 0.0140 [0.0109, 0.0177], versus the Wilks expectation 0.0146. The selection-calibrated maximum-influence result is reported in §5; its conditional result shows that localization should not be counted as independent evidence against ΛCDM. Full CAMB computation (a separate execution environment, not scientific independence) confirms the baseline and LRG2-deletion fits to approximately0.2 in $\Delta\chi^2$. Posterior sampling under the official priors shows no pileup at the $w_0 + w_a < 0$ boundary; the naive mean-plug-in DIC convention (-8.99) is documented as an artifact of the curved w_0--w_a ridge, with the standard MAP plug-in giving -4.96.

IX. PRE-REGISTERED CROSS-PROBE CONSISTENCY TEST (H-X)

With hypotheses and kill conditions registered before execution (HYPOTHESIS-cross-probe.md), per-probe $w_0 w_a$ posteriors were sampled for P1 = DESI BAO+cCMB, P2 = Pantheon+ +cCMB+BBN, P3 = DES5Y+cCMB+BBN. Pairwise 2D parameter-shift tensions: P1–P2 = 1.44σ, P1–P3 = 1.35σ, P2–P3 = 0.24σ. Phantom-crossing redshifts: P1 z× = 0.487 [0.418–0.553]; P2 0.292 [0.187–0.502]; P3 0.296 [0.244–0.403]. Verdict against the registered conditions: INTERMEDIATE — neither the consistency (<1σ all pairs) nor inconsistency (>2σ any pair) threshold met; reported without promotion. The registered follow-up is complete: P1 without LRG2 gives z× = 0.478 [0.384–0.564], so LRG2 controls much of the preference's strength but not the surviving BAO posterior's crossing location. Caveats: Gaussian shift metric on curved posteriors; surrogate likelihood (CAMB-validated); shared-anchor correlation.

X. THE FULL-CMB LENSING VARIANT

Using cobaya with CAMB and the clik-free native Planck 2018 likelihoods (low-ℓ TT, low-ℓ EE, plik-lite TTTEEE, lensing) plus the DESI BAO likelihood, MAP minimization (BOBYQA, dual converged runs) under DESI's official w(z) priors gives, after removing the uniform-prior volume constant: with LRG2, $\Delta\chi^2$ = -10.35 (2.77σ); without LRG2, $\Delta\chi^2$ = -3.76 (1.43σ). The with-LRG2 value sits below the published PR4-based -12.5 (3.1σ) by an amount consistent with the PR3-vs-PR4 likelihood difference (pre-registered caveat). The central finding stands at the full-likelihood level: **CMB lensing does not substitute for LRG2** — the preference collapses from ~2.8σ to ~1.4σ on removal of one measurement. Caveats: PR3-native likelihoods; MAP not posterior comparison; single pipeline pending clean-room rerun; the selected-null calibration and its conditional limit are in §5.

XI. WHAT THIS ESTABLISHES — AND WHAT IT DOES NOT

Established: the published compressed likelihoods genuinely contain the reported preference; it survives independent reimplementation, optimizer attack, full Boltzmann physics, parameterization changes, and bootstrap calibration. Within CPL, a direct constant-w null test gives a calibrated 2.75σ preference for time variation, falling to 1.81σ without LRG2. The held-out LRG2 vector is a modest 2.05σ discrepancy, driven more by isotropic distance scale than by its shape ratio. Among ΛCDM mocks with a global preference this strong, a maximum deletion influence at least as large as observed occurs about half the time. Not established: that dark energy evolves; that LRG2 is wrong; that localization or the targeted held-out p-value is independent evidence; that the supernova calibration explanation is correct; or that the PR3-native full-CMB variant reproduces DESI's PR4/ACT stack. The honest status is **a reproducible, LRG2-sensitive 2–3σ hint — not proof.**

XII. FRONTIER-STANDARD BOUNDARY AND DEVELOPMENT ROADMAP

A frontier-roadmap review was compared against the frozen analysis rather than treated as new evidence. Several recommendations are adopted as the next evidentiary gates: native-likelihood reconstruction of LRG2; survey splits and nuisance templates for fiber assignment, reconstruction, redshift failures, observing conditions, covariance, and window functions; flexible or nonparametric expansion reconstructions calibrated inside null mocks; prior-sensitive Bayesian evidence or out-of-sample predictive scores; growth consistency through RSD, $f\sigma_8$, weak lensing, and CMB lensing; and a hierarchical, survey-aware supernova calibration model.

The same review also proposed interpretations that exceed this project's evidence. This compressed-likelihood analysis does not identify fiber assignment as the cause of LRG2, prove that the supernova seam is a calibration error, validate a Gaussian-process crossing, support Early Dark Energy, or favor interacting/decaying dark matter or modified gravity. Those ideas remain hypotheses until they improve predictions outside the data that generated the anomaly and survive appropriate null calibration and model-complexity penalties. The complete comparison is recorded in `FRONTIER_REVIEW.md`.

The frontier claim ladder is therefore explicit:

- **Reproducible likelihood feature: Required evidence:** independent implementation and stable optimization; **Current status: Met** for the published compressed inputs
- **Calibrated anomaly: Required evidence:** null simulations and selection-aware statistics; **Current status: Met** for the global, maximum-influence, direct-time-variation, and held-out tests
- **Measurement diagnosis: Required evidence:** native likelihood, survey splits, nuisance-template and covariance audit; **Current status: Open**
- **Cross-probe physical consistency: Required evidence:** expansion, growth, and lensing predicted by one model; **Current status: Open**
- **Model preference: Required evidence:** prior-robust Bayesian or predictive advantage over flexible alternatives; **Current status: Open**
- **Discovery: Required evidence:** independent data and human replication at discovery-grade significance; **Current status: Not met**

XIII. FALSIFIABLE EXPECTATIONS

If the signal is physics: DESI's final likelihood (2027) retains the z approximately 0.7 distance-scale departure at higher precision; Rubin-calibrated supernovae reproduce the low-z crossing with no magnitude offset required; growth and lensing co-vary as dynamical dark energy predicts. If it is systematics: the LRG2 distance residual regresses; the supernova preference tracks calibration choices across compilations; residuals decorrelate across probes. We pre-commit to updating this document against those outcomes.

XIV. REPRODUCTION

The blind reproduction protocol and one-command runners are documented in clean-room/README.md. Released outputs are in results/; canonical values, completion states, and checksums are in RESULTS_MANIFEST.json.

XV. ACKNOWLEDGMENTS AND PROVENANCE OF ASSISTANCE

Implementation, literature triage, and drafting were AI-assisted (multiple systems, adversarially cross-checked); four AI software/statistics/manuscript audits are incorporated with attribution in the README, but these are not independent scientific replication. A human clean-room rerun remains pending. Errors are the author's. Data credits: DESI, SDSS/eBOSS, Pantheon+/SH0ES, DES, Planck, and the cobaya project's data releases.

XVI. PRIMARY SOURCES

1. DESI Collaboration, "DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints," arXiv:2503.14738.
2. DESI Collaboration, "DESI 2024 III: Baryon Acoustic Oscillations from Galaxies and Quasars," arXiv:2404.03000, and "DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations," arXiv:2404.03002.
3. Brout et al., "The Pantheon+ Analysis: Cosmological Constraints," arXiv:2202.04077.
4. DES Collaboration, "The Dark Energy Survey: Cosmology Results With ~1500 New High-redshift Type Ia Supernovae Using the Full 5-year Dataset," arXiv:2401.02929.
5. Efstathiou, "Evolving Dark Energy or Supernovae Systematics?", arXiv:2408.07175.
6. Vincenzi et al., "Comparing the DES-SN5YR and Pantheon+ SN cosmology analyses," arXiv:2501.06664.
7. Cortês and Liddle, "Interpreting DESI's evidence for evolving dark energy,"

arXiv:2504.15336.

DATA AVAILABILITY

The released code, download instructions, result files, chains, and registered hypothesis records are available at `https://github.com/GrobeStreet/de-stress-lab`. The immutable flagship scientific archive is doi:10.5281/zenodo.21632602. The canonical completion states and checksums are recorded in `RESULTS_MANIFEST.json`. A human clean-room execution remains pending.

CONFLICT OF INTEREST

The author declares no competing financial interests.